

Basic Mathematics

Mathematics

Lecture Notes

A friendly path through algebra, geometry, and calculus.

June 6, 2026

Contents

1	Introduction	2
2	Linear Algebra	4
2.1	Vectors	4
2.2	Matrices	4
2.3	Systems of Linear Equations	4
3	Analytic Geometry	5
3.1	Coordinates	5
3.2	Lines	5
3.3	Planes	5
4	Calculus	6
4.1	Functions	6
4.2	Limits	6
4.3	Continuity	6
4.4	Derivatives	6

Chapter 1

Introduction

These notes are an invitation to look at mathematics in a different way. For many students, mathematics has long appeared as a collection of formulas, rules, and mnemonic patterns: one recognizes the type of problem, chooses a remembered procedure, performs the required operations, and moves on. Such skills have their place, but they are not the heart of mathematics. Mathematics begins when we stop for a moment and ask what we are really looking at, what kind of object has appeared, what question is being asked, what is known, what is unknown, and why a proposed step is justified.

The purpose of this course is therefore not only to practise calculations. We will learn how to read mathematical expressions, how to formulate precise questions, how to distinguish an object from its notation, how to notice structure, and how to build simple but meaningful chains of reasoning. A solved problem is not valuable only because it produces an answer. It is valuable because it can reveal a method of thought: how to begin, what to observe, which assumptions matter, what can be transformed, what remains unchanged, and how a conclusion follows from earlier facts.

The course will focus on three fundamental parts of mathematics: linear algebra, analytic geometry, and calculus. Linear algebra introduces vectors, matrices, systems of equations, and transformations. Analytic geometry shows how algebraic equations describe points, lines, planes, directions, distances, and angles. Calculus gives a language for functions, limits, continuity, derivatives, integrals, change, and accumulation. These topics are not isolated chapters placed next to one another. Together they form a basic language for describing structure, space, and change.

We will try to study this language from the inside. A definition should not be treated as a sentence to memorize, but as a way of introducing a new kind of object. A theorem should not be treated as a decorative statement, but as the result of asking a clear question under precise assumptions. An example should not be treated as a template to copy blindly, but as a small experiment in reasoning. Even a simple calculation can show something important if we ask why it works and what it tells us.

For this reason, some arguments will deliberately return in different places. The same habit of asking what is fixed and what is variable appears when we study functions, vectors, equations, and transformations. The same need for precision appears when we define a line, compute a derivative, solve a system of equations, or interpret an integral. The same mathematical question can often be seen algebraically, geometrically, and conceptually. This repetition is intentional: it helps the reader recognize patterns of reasoning rather than only patterns of exercises.

The number of solved tasks is not the main measure of progress. What matters is how the tasks are solved. Did we understand the question? Did we identify the object correctly? Did we choose notation consciously? Did we know why each step was allowed? Did we pause to interpret the result? Did the calculation teach us something about the idea behind it? These questions are part of the work. They are not an addition to mathematics; they are mathematics being done carefully.

Mathematics is useful in physics, engineering, computer science, data analysis, modelling, and many other fields, but its value is not only practical. It is also a way of seeing. It teaches precision, discipline, imagination, and the ability to move from simple observations to general conclusions. These notes are meant to show a piece of that beauty, strength, and richness. The goal is not merely to survive mathematical techniques, but to begin to appreciate mathematics

as an art of asking good questions and giving well-reasoned answers.

Chapter 2

Linear Algebra

This chapter introduces the basic objects and methods of linear algebra.

2.1 Vectors

This section will introduce vectors and their basic operations.

2.2 Matrices

This section will introduce matrices and basic matrix operations.

2.3 Systems of Linear Equations

This section will introduce systems of linear equations and their connection with matrices.

Chapter 3

Analytic Geometry

This chapter introduces geometric objects described using coordinates and algebraic equations.

3.1 Coordinates

This section will introduce coordinate systems and the representation of points.

3.2 Lines

This section will introduce equations of lines and their geometric interpretation.

3.3 Planes

This section will introduce equations of planes in three-dimensional space.

Chapter 4

Calculus

This chapter introduces functions, limits, continuity, and derivatives.

4.1 Functions

This section will introduce functions as mathematical objects describing dependence between quantities.

4.2 Limits

This section will introduce the concept of a limit.

4.3 Continuity

This section will introduce continuity and its basic interpretation.

4.4 Derivatives

This section will introduce derivatives as rates of change.